1 Assignment 4a — Neutrino Oscillations in Matter

The neutrino time evolution in matter is governed by Schrödinger equation with an effective Hamiltonian that includes both the vacuum term and a matter potential term. In class, we saw that we can write the time evolution as

$$i\frac{d}{dt}\begin{pmatrix}\nu_e \\ \nu_\mu\end{pmatrix} = \left[U\begin{pmatrix}E_1^0 & 0 \\ 0 & E_2^0\end{pmatrix}U^\dagger + \begin{pmatrix}V_{CC} & 0 \\ 0 & 0\end{pmatrix}\right]\begin{pmatrix}\nu_e \\ \nu_\mu\end{pmatrix}, \tag{1.1}$$

where $E_i^0 \simeq p + \frac{m_i^2}{2p}$ are the energy eigenvalues in vacuum, $U = R(\theta) \equiv \begin{pmatrix}\cos\theta & \sin\theta \\ -\sin\theta & \cos\theta\end{pmatrix}$, and V_{CC} is the charged-current matter potential.

(**4 points**) 1. Define the effective mixing angle in matter, θ_m, such that the effective Hamiltonian can be diagonalized as

$$\hat{H}_m = U_m\begin{pmatrix}E_1^m & 0 \\ 0 & E_2^m\end{pmatrix}U_m^\dagger, \quad \text{where} \quad i\frac{d}{dt}\nu_i^m = E_i^m\nu_i^m, \tag{1.2}$$

where $U_m = R(\theta_m)$ is the mixing matrix with angle θ_m that takes us to the diagonal basis in matter $\nu_{\text{flavor}} = U_m\nu^m$. The script m indicates that these are the matter eigenstates and eigenvalues. We have implicitly assumed that the matter potential is constant throughout the neutrino propagation (take $t \simeq x$ and Eq. (1.1) above tells you how the state evolves as it propagates from source to detector). Derive an expression for the eigenvalues E_i^m and $\sin 2\theta_m$ in terms of the vacuum mixing angle θ, the neutrino energy E_ν, the mass-squared difference $\Delta m^2 = m_2^2 - m_1^2$, and the matter potential V_{CC}. Define a matter-basis mass splitting as $\Delta m_m^2/(2E_\nu) \equiv E_2^m - E_1^m$ and write your expression for Δm_m^2 as a function of these parameters.
(Hint: Recall that diagonal terms in the time evolution will not contribue to oscillations.)

(Hint 2: Recall also that a real symmetric 2×2 matrix $\begin{pmatrix}a & c \\ c & b\end{pmatrix}$ can be diagonalized by an orthogonal transformation with angle θ such that $\tan 2\theta = \frac{2c}{b-a}$ or $\sin 2\theta = \frac{2c}{\sqrt{(b-a)^2+4c^2}}$.
(Hint 3: Your answer should look like $\sin^2 2\theta_m = \sin^2 2\theta/R$ and $\Delta m_m^2 = \Delta m^2\sqrt{R}$ for some R factor.)

(**2 points**) 2. Using the expression derived in Question 1, find the condition for which the effective mixing angle in matter θ_m becomes maximal (i.e., $\theta_m = \pi/4$). What does this mean for oscillations? This condition is known as the Mikheyev-Smirnov-Wolfenstein (MSW) resonance condition.

(**2 points**) 3. Given that $V_{CC} = \sqrt{2}G_F N_e$, where G_F is the Fermi constant and N_e is the electron number density in matter, estimate the neutrino energy E_ν at which the MSW reso-

nance occurs for solar neutrinos propagating through the Sun
(Hint: take $N_e \sim 100 N_A$ cm^{-3}, where N_A is Avogadro's number. Assume $\Delta m^2 = 7.5 \times 10^{-5}$ eV^2 and $\theta = 33°$.)

(**2 points**) 4. The survival probability in matter becomes $P(\nu_e \to \nu_e) = 1 - \sin^2 2\theta_m \sin^2\left(\frac{\Delta m_m^2 t}{4E_\nu}\right)$, analogous to the vacuum oscillation probability. What happens when the MSW resonance condition is exactly satisfied throughout the entire neutrino propagation for $\Delta m^2 t/2E_\nu \gg 1$? What about when $\Delta m_M^2 t/2E_\nu \ll 1$ (expand the sine functions)?

(**Bonus 2 points**) 5. For a changing matter density, the effective mixing angle θ_m also changes along the neutrino trajectory. Derive the adiabaticity condition for which a neutrino that starts as a matter eigenstate ν_1^m (or ν_2^m) remains in that eigenstate throughout the propagation. What happens when this condition is satisfied?